

mathematical reasoning (S. Lee et al., 2024; McCoy et al., 2024; Mirzadeh et al., 2024), even when they mimic or surpass human performance in other ways. For example, larger models improve in certain domains (e.g., some challenging pattern recognition tasks) but still fail at seemingly simple tasks that humans would expect them to handle easily (Zhou et al., 2024). The depth of understanding gained without interactive, multisensory embodiment may differ from human cognition (Jones & Bergen, 2024), resembling, to some extent, the difference between inferring object properties from descriptions or observations versus learning through direct manipulation and sensory feedback, where crucial experiential nuances can be missed (cf. Kim et al., 2019, regarding visual properties and congenital blindness).

Paradoxically, solutions to these conceptual limitations may render future AI systems *less* analogous to human cognition even while they gain superhuman abilities. For instance, addressing grounding limitations through embodied robotics with RL would introduce fundamentally different learning processes from both current text-based training and human development—such as active, continuous learning from direct experience via grounded rewards from the world using non-human tools, rather than relying on human input. Similarly, systems designed to overcome reasoning limitations might develop internal representations and goals with no human analogue. For example, as relentless problem-solving agents, these systems may exhibit a distribution of competence that differs from that of an average person.

As the field progresses beyond current paradigms, the relationship between machine and human cognition will continue to evolve, requiring ongoing reassessment of the role of LLMs in psychological science.

The Average Human Fallacy

Mischaracterizing the nature of intelligence in LLMs may not matter as much if they behave or perform like the average human, enabling them to functionally replace human participants. But this assumption commits the average human fallacy, which misrepresents what LLM outputs signify.

Consider the engineering purpose and approach of LLMs. They are explicitly developed to outperform humans, as measured by a broad range of benchmarks and standardized tests—an engineering feat bolstered by continually improved design and algorithms, and access to more data and compute, while free from biological limitations (see also **Box 2**). In actual tests, for example, GPT-4 outperforms humans in detecting and interpreting irony, recognizing indirect requests or hints in conversation (Strachan et al., 2024), analogical reasoning tasks (Webb et al., 2023), and probabilistic reasoning tasks like the Linda/Bill problems and the bat-and-ball problem (Yax et al., 2024), but underperforms humans in tasks like faux pas tests (Strachan et al., 2024). This contrasts with the assumption of the replacement view that LLM responses mirror average human judgments from the training data (Dillion et al., 2023) or the majority’s mainstream opinions (Qu et al., 2024).

The generalizability challenge is multifaceted. First, LLMs are not representative samples of any defined human population, let alone an “average” human. Their training data (mostly online text) exhibit systematic biases—predominantly Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations, especially those who are hegemonic, young, and publicly expressive (Crockett & Messeri, 2023; Santurkar et al., 2023; Tao et al., 2024). This biased training creates a complex landscape for psychological simulation: While LLMs might adequately capture behaviors that are largely universal across human cultures (certain basic cognitive or emotional phenomena, perhaps), they face

limitations when simulating those known to vary across cultures and groups—from number representations to personality traits and moral reasoning. Further complicating this picture is our incomplete understanding of which psychological phenomena are universal versus culturally variable. This uncertainty means researchers must exercise particular caution when using WEIRD-biased LLMs—or any LLMs for that matter—to simulate potentially culture-dependent phenomena.

In tandem with spatial biases, the temporal distribution of the data is more concentrated in recent history, with the model’s understanding of the past filtered through the lens of contemporary languages and norms (Ziems et al., 2024). This presentist and recency bias risks temporal flattening, wherein historical and contemporary voices are homogenized, obscuring the richness of historical diversity in human thought. This is problematic for probing thoughts and behaviors that have evolved over extended periods—such as the way people think about concepts like gender, race, and class (Kozlowski & Evans, 2024). The models may lack the contextual richness and temporal granularity needed to understand how these concepts were discussed in different periods, potentially reinforcing contemporary biases when trying to understand people from earlier time periods.

Beyond training data biases, the engineering goals of LLMs further compromise their representational accuracy. The goal of developing advanced LLMs—to provide accurate, helpful answers to users—contrasts with the study of human cognition, which is replete with inaccuracies, biases, shortcuts, and idiosyncrasies. For instance, in language comprehension, people often settle for a partial and sometimes inaccurate understanding that is nevertheless sufficient for the task at hand—“good-enough representations” (Ferreira et al., 2002). In decision making, when emotions are induced in bargaining games and repeated cooperation games, GPT-4 tends to maintain consistent, rational decision-making, in contrast to humans (Mozikov et al., 2024).

Due to model limitations, opacity, and the common use of RLHF, such models cannot be assumed to represent the “average” of their training data. Responses may reflect hallucination, the influence of RLHF, or other bias reduction efforts; they could also simply regurgitate specific instances, examples, or strategies from their training data, particularly when the query is well represented in the data (Aher et al., 2023; Binz & Schulz, 2023a; Shiffrin & Mitchell, 2023). For example, when LLM responses are altered through RLHF, they can deviate from the original training data—and may even exacerbate their misalignments with non-dominant views (Santurkar et al., 2023). In this process, the communicative intent of developers (and data labelers) shapes model outputs to prioritize goals like accuracy and helpfulness, rather than reflecting “raw” human-like responses from the training texts.

This raises concerns about whether the outputs genuinely represent human responses. If the psychological construct being measured is meant to be “average human response,” the LLM fails to operationalize this construct due to the representational biases described above. The models often become “too neutral, detached, and nonjudgmental,” lacking selfhood and initiative (Ye et al., 2024), and exhibiting homogeneous personality profiles—high in agreeableness and low in neuroticism (Li & Qi, 2025; Pellert et al., 2024). While this fine-tuning process can reduce certain biases (Hu et al., 2025), enhance accuracy, and make the model more pleasant to interact with, it also weakens its ability to reflect the actual attitudes and thoughts present in human texts (Harding et al., 2024) and to produce diverse responses (Murthy et al., 2024). Such alterations may also introduce new preferences or biases from the feedback, including sycophancy (a tendency to generate outputs that are excessively user-pleasing or conform to perceived desirable responses), making it challenging to rely on RLHF-tuned LLMs as accurate indicators of human thought and judgment (Park et al., 2024). In survey responses, for example, LLMs do not share human response biases, with more

pronounced discrepancies in RLHF-tuned models (Tjauatja et al., 2024). Likewise, within the GPT-3 family, fine-tuned models exhibited higher propensity for conjunction fallacy and intuitive reasoning relative to base models (Yax et al., 2024).

The practical implications of these misalignments are evident in actual response patterns. LLM responses have been found to mischaracterize marginalized groups, as evidenced by out-group imitation rather than in-group description (Wang et al., 2025), and misrepresent sampled groups, as demonstrated by upward bias in mean ratings of the Big Five personality traits (Niszczoła et al., 2025), as well as in other surveys and tests (Hagendorff et al., 2023; Sarstedt et al., 2024; Tjauatja et al., 2024). In addition to these shifts in average responses, LLMs also fail to capture the nuances and heterogeneity of human responses, producing flattened, oversimplified portrayals of various groups (Wang et al., 2025).

While diversifying the training corpus to include more languages and cultural contexts helps to broaden representations, achieving global representations is ultimately a long-term challenge (Lin & Li, 2023). In the foreseeable future, the quantity and quality of available non-English training texts will remain impoverished compared to English as the *lingua franca*.

To mitigate the average human fallacy, key considerations include explicitly defining represented populations, potentially using specialized models, testing prompts and models for consistency and bias, validating responses against human data, and transparently reporting representational limitations. Specifically:

1. **Model selection and customization:** Be explicit about which populations are represented in the training data and how this affects generalizability claims. Consider using specialized models fine-tuned on representative human data and, importantly, verify performance with actual samples from those populations (Gao et al., 2024; Suh et al., 2025).
2. **Prompt design:** Test models across multiple prompting styles to assess the consistency of simulated responses and determine sensitivity to superficial variations (Li & Qi, 2025; Momennejad et al., 2023; Tao et al., 2024). Compare outputs from different LLM versions and architectures to identify systematic biases in representation.
3. **Interpretations and applications:** Validate LLM responses against human samples, particularly for claims about specific populations or demographic groups, and report cases where LLM responses diverge from human patterns. Avoid generalizing beyond domains where empirical validation has been established (Abdurahman et al., 2025).
4. **Ethics:** Document potential biases and acknowledge limitations in representativeness when reporting results from LLM simulations (Abdurahman et al., 2025).

Alignment as Explanation Fallacy

High output alignment between LLM responses and human data has been suggested as evidence that LLMs can sometimes replace human participants (Dillion et al., 2023). While such alignment suggests that LLMs might capture something mechanistic about human behavior, it is fallacious to assume mechanistic equivalence—that the model explains human cognition (Guest & Martin, 2023) or can replace human participants for theoretical understanding (see **Box 4**). Treating such alignment as sufficient for mechanistic equivalence constitutes the alignment-as-explanation fallacy. This fallacy arises when output similarity is conflated with representational and processing similarity—that is, when similar outputs are taken as evidence that LLMs engage the same psychological constructs or cognitive mechanisms as humans, overlooking the differences between the statistical pattern-matching of current LLMs and the embodied, contextual cognitive processes of humans.